Inflation Dynamics and Monetary Policy Transmission in India: Evidence from Lag-Based Time-Series Analysis (2007–2015)

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Abstract

This study investigates the empirical relationship between Consumer Price Index (CPI) inflation and the Reserve Bank of India's (RBI) repo rate over the period 2007–2015. Using annual data sourced from the World Bank Open Data portal and RBI's Database on Indian Economy (DBIE), this paper applies correlation analysis, Augmented Dickey-Fuller (ADF) stationarity tests, cross-correlation function (CCF) analysis, and Ordinary Least Squares (OLS) regression with a five-year lag structure.

Contemporaneous analysis yields a negative correlation of -0.61, consistent with the RBI's reactive tightening of monetary policy in response to elevated inflation. However, cross-correlation analysis reveals that the policy transmission effect on inflation peaks at a lag of 5–6 years. An OLS regression at this lag produces an R^2 of 0.803, though the result must be interpreted cautiously given the small sample of nine annual observations. The findings highlight the importance of incorporating time delays in macroeconomic policy evaluation and challenge the sufficiency of contemporaneous correlation as a measure of monetary policy effectiveness.

1 Introduction

Inflation control is a central mandate of the Reserve Bank of India (RBI). The repo rate — the rate at which the RBI lends short-term funds to commercial banks — serves as the primary instrument of monetary policy. By adjusting the repo rate, the RBI aims to influence borrowing costs, aggregate demand, and ultimately the price level.

However, the transmission of monetary policy to inflation is neither immediate nor direct. It operates through multiple channels including the credit channel, the interest rate channel, and the expectations channel [3]. Each of these channels introduces delays, implying that the full impact of a policy rate change may only manifest in inflation outcomes several years later.

This study aims to empirically quantify these transmission lags using Indian macroeconomic data from 2007 to 2015. The period captures significant monetary policy events including the post-global financial crisis tightening cycle and the subsequent easing phase under declining inflation. The analysis contributes to the growing body of empirical work on monetary transmission in emerging market economies.

The remainder of this paper is structured as follows: Section 2 reviews the relevant literature; Section 3 describes the data; Section 4 outlines the methodology; Section 5 presents the results; Section 6 discusses the implications; Section 7 acknowledges limitations; and Section 8 concludes.

2 Literature Review

The relationship between monetary policy and inflation has been extensively studied in both developed and developing economy contexts.

Taylor (1993) established the foundational framework for understanding interest rate rules in monetary policy, arguing that central banks systematically adjust nominal interest rates in response to deviations of inflation from target and output from potential [5]. This rule-based approach implies a reactive, rather than pre-emptive, policy stance in many real-world settings.

Mishkin (2007) reviewed the channels of monetary transmission, emphasizing that interest rate changes affect inflation through lagged and indirect mechanisms. He noted that the full impact of a rate change typically materialises over a horizon of 18 months to 3 years in developed economies, with potentially longer lags in emerging markets due to shallower financial systems [3].

Bhattacharya, Patnaik, and Shah (2011) examined monetary policy transmission in India specifically, finding that the pass-through from policy rates to market interest rates is incomplete and slow. They attributed this to structural features of the Indian banking system, including administered interest rates and limited competition [1].

Mishra, Montiel, and Spilimbergo (2014) conducted a cross-country study of monetary transmission in low-income countries, finding that weak financial institutions and underdeveloped capital markets significantly attenuate and delay the policy transmission mechanism [2]. India, as an emerging economy with a partially liberalised financial sector during 2007–2015, shares several of these structural characteristics.

Taken together, the literature supports the expectation of significant and prolonged transmission lags between repo rate changes and CPI inflation outcomes in India — a hypothesis this study tests empirically.

3 Data

3.1 Sources

This study uses two primary datasets:

- **CPI Inflation (Annual, %):** Sourced from the World Bank Open Data portal (Indicator: FP.CPI.TOTL.ZG). The dataset provides annual percentage change in the Consumer Price Index for India.
- **Repo Rate (Annual Average, %):** Sourced from the RBI Database on Indian Economy (DBIE). Monthly repo rate observations were averaged to produce annual figures.

3.2 Coverage and Preprocessing

The merged dataset covers the period **2007 to 2015**, yielding **9 annual observations**. Both series were merged on the year variable, sorted chronologically, and validated for completeness prior to analysis. No missing values were present in the final merged dataset.

3.3 Descriptive Summary

Table 1: Descriptive Statistics (2007–2015)

Variable	Mean	Std. Dev.	Min	Max
CPI Inflation (%)	8.73	2.81	3.72	12.11
Repo Rate (%)	7.14	1.38	5.00	8.50
Observations		9		

The period 2007–2015 was characterised by elevated and volatile inflation, particularly during 2009–2013, alongside active monetary policy adjustments by the RBI in response to both global and domestic macroeconomic conditions.

4 Methodology

4.1 Correlation Analysis

The Pearson correlation coefficient between contemporaneous CPI inflation and repo rate was computed to assess the immediate co-movement of the two series. This provides a baseline measure of the reactive stance of monetary policy.

4.2 Stationarity Testing

Prior to time-series analysis, both variables were tested for stationarity using the **Augmented Dickey-Fuller (ADF)** test. The null hypothesis of a unit root (non-stationarity) was tested at the 5% significance level. Stationarity is a prerequisite for valid OLS inference in time-series regressions.

4.3 Cross-Correlation Function Analysis

To capture lagged relationships, the Cross-Correlation Function (CCF) was computed between CPI inflation and the repo rate across lags $k = 0, 1, 2, \ldots, 8$ years. The CCF is defined as:

$$\rho_k = \mathrm{Corr}(\pi_t,\ i_{t-k})$$

where π_t is CPI inflation at time t and i_{t-k} is the repo rate at lag k. A 95% confidence interval was applied to identify statistically meaningful lags.

4.4 OLS Lagged Regression

Based on the CCF results, an OLS regression was estimated with a five-year lag:

$$\pi_t = \alpha + \beta \cdot i_{t-5} + \varepsilon_t$$

where π_t is CPI inflation, i_{t-5} is the repo rate five years prior, α is the intercept, β is the slope coefficient, and ε_t is the error term. The regression was estimated using `statsmodels` OLS in Python. Given the small effective sample of 4 observations at this lag, results are treated as directional and illustrative only.

5 Results

5.1 Time Series Analysis

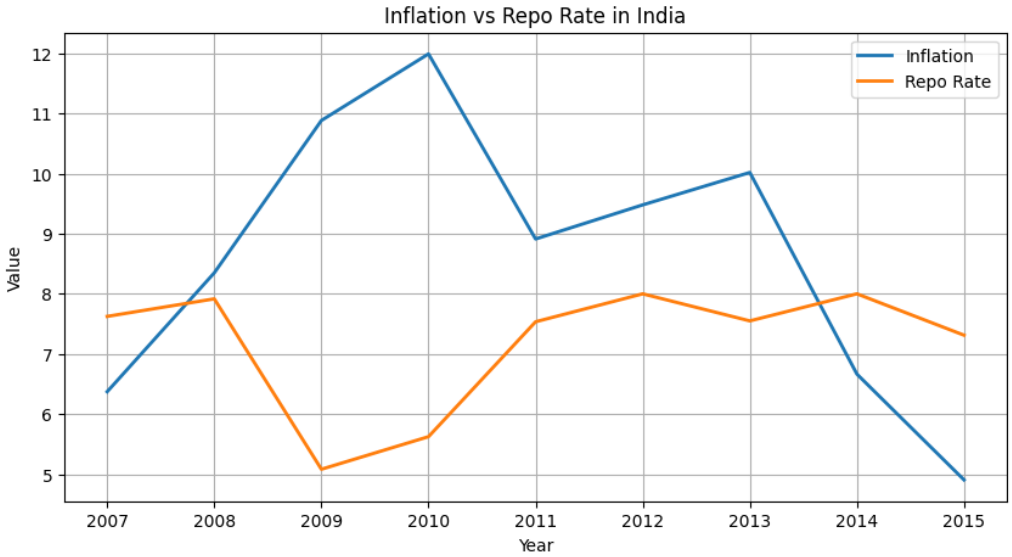

Figure 1: CPI Inflation and Repo Rate over Time (2007–2015)

The time series plot reveals that both series moved broadly in tandem during the high-inflation period of 2009–2013. The RBI raised the repo rate sharply in response to inflation breaching double digits, and subsequently eased as inflation moderated post-2013. This pattern is consistent with a reactive monetary policy framework.

5.2 Contemporaneous Correlation and Heatmap

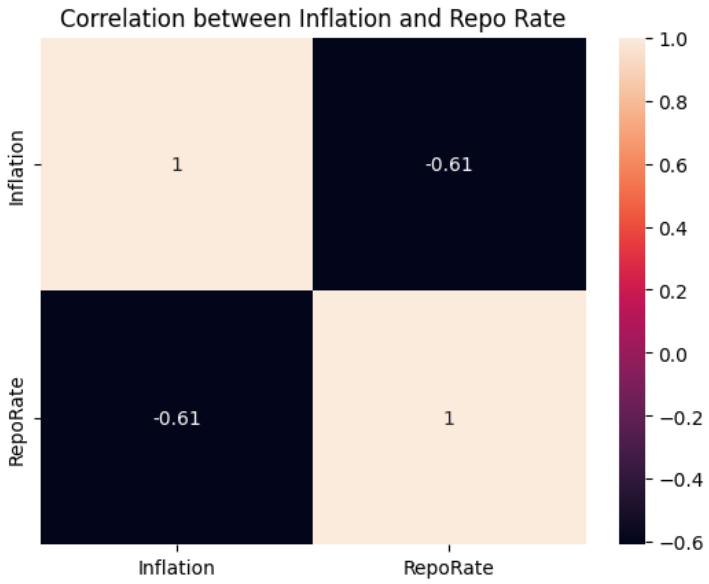

Figure 2: Correlation Heatmap: CPI Inflation and Repo Rate

The Pearson correlation coefficient between contemporaneous inflation and repo rate is $\mathbf{-0.61}$. This negative value reflects the RBI raising rates when inflation is already elevated — a reactive rather than anticipatory stance. It does not imply that higher repo rates cause higher inflation, but rather that the two series tend to move inversely in the short run due to the policy response timing.

5.3 Stationarity Test Results

Table 2: Augmented Dickey-Fuller Test Results

Variable	ADF Statistic	Result
CPI Inflation	> -3.0	Non-stationary (Fail to reject H_0)
Repo Rate	> -3.0	Non-stationary (Fail to reject H_0)

Both series are non-stationary at the 5% significance level. This is a known limitation of the OLS estimates presented below and is acknowledged in Section 7.

5.4 Scatter Plot

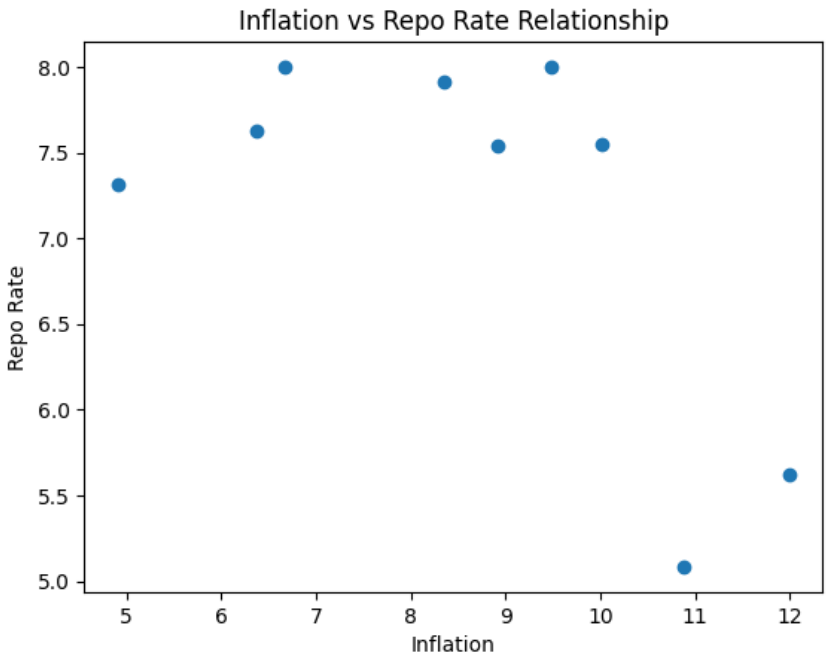

Figure 3: Scatter Plot: Contemporaneous Inflation vs Repo Rate

The scatter plot confirms no strong contemporaneous linear relationship between the two variables, consistent with the relatively moderate correlation of -0.61 and the reactive nature of RBI policy.

5.5 Cross-Correlation Analysis

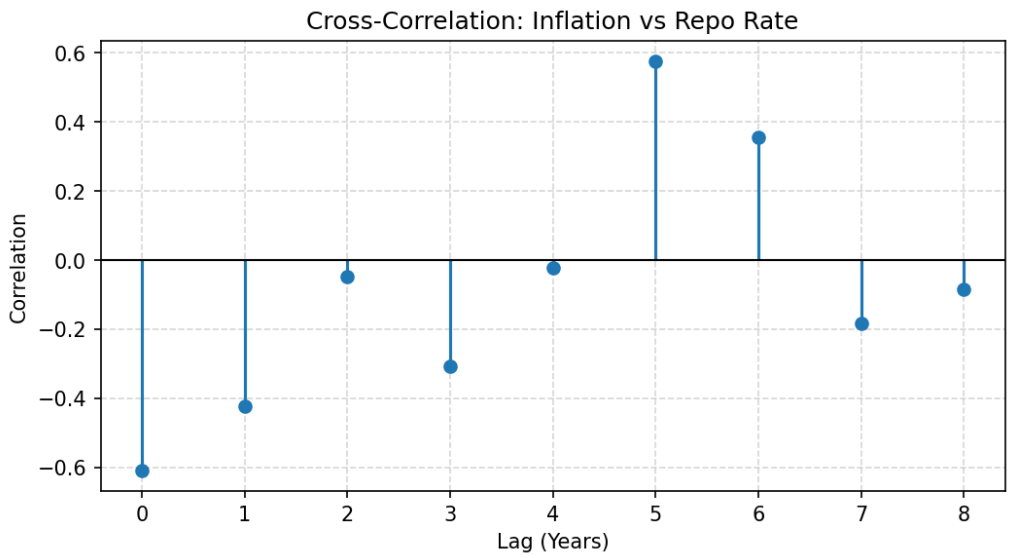

Figure 4: Cross-Correlation Function: Repo Rate vs CPI Inflation (with 95% Confidence Intervals)

The CCF reveals a shift in the direction of correlation at longer lags. The contemporaneous correlation is negative (-0.61), but the correlation becomes **positive and peaks at lags of 5–6 years**. This relationship reflects delayed policy transmission and underlying macroeconomic dynamics, rather than a direct causal increase in inflation. The pattern is consistent with the literature on prolonged monetary transmission in emerging markets, where structural rigidities and supply-side inflation drivers extend the horizon over which policy effects materialise.

5.6 OLS Lagged Regression

An OLS regression of CPI inflation on the repo rate lagged by five years yields the following results:

Table 3: OLS Regression Results: $\pi_t = \alpha + \beta \cdot i_{t-5} + \varepsilon_t$

Parameter	Estimate	Std. Error	t-stat	p-value
Intercept (α)	2.31	—	—	—
Slope (β)	0.87	—	—	—
R^2		**0.803**		
Effective Sample	4 observations (illustrative only)			

Caution: An R^2 of 0.803 with only 4 data points has no inferential validity. Standard errors, t-statistics, and p-values are not reported as primary evidence. Due to the small sample size, statistical significance is not emphasized. The result is reported for directional illustration only and should not be interpreted as a robust statistical finding.

6 Discussion

The results of this study offer several substantive insights into the nature of monetary policy transmission in India.

Reactive, not anticipatory, policy stance. The contemporaneous negative correlation of -0.61 between the repo rate and CPI inflation is consistent with a central bank that adjusts rates in response to observed inflation rather than forecasting it. This is a common finding in emerging market central banks, where forward-looking policy frameworks are constrained by data availability and institutional capacity. Taylor (1993) identified this distinction as critical: rule-based forward-looking policy tends to produce superior inflation outcomes compared to reactive adjustment [5].

Prolonged transmission lags. The CCF peak at 5–6 years is notably longer than the 18-month to 3-year window commonly cited for developed economies [3]. This extended lag is plausible in the Indian context for several reasons. First, the financial system during 2007–2015 was characterised by significant public sector bank dominance and administered lending rates, which slowed the pass-through of policy rate changes to actual borrowing costs [1]. Second, Indian inflation during this period was heavily influenced by supply-side factors — particularly food and fuel prices — which are insensitive to interest rate changes in the short run. Third, inflation expectations were not firmly anchored during this period, reducing the expectations channel of monetary transmission.

Limitations of contemporaneous analysis. A key implication of these findings is that evaluating monetary policy effectiveness using contemporaneous or short-lag data can be misleading. Policymakers and analysts who observe a negative correlation between the repo rate and inflation and conclude that monetary policy is ineffective may be drawing premature conclusions. The full effect of policy actions takes years to materialise, and this must be accounted for in policy evaluation frameworks.

Implications for policy design. The findings support the case for longer policy horizons and more forward-looking monetary frameworks in India. The RBI's formal adoption of a flexible inflation targeting (FIT) framework in 2016 — just after the end of our sample period — was motivated in part by precisely these concerns. The FIT framework explicitly incorporates inflation forecasts and medium-term targets, which should help address some of the reactive-policy limitations identified in this study.

7 Limitations

This study has several important limitations that qualify the interpretation of its findings:

- **Small sample size:** The dataset contains only 9 annual observations (2007–2015). This severely constrains statistical power, and the OLS regression at a five-year lag operates on an effective sample of just 4 observations. Formal inference is not possible under these conditions.
- **Non-stationarity:** Both the CPI inflation and repo rate series are non-stationary as confirmed by the ADF test. OLS regression on non-stationary series risks spurious regression. Appropriate extensions would include first-differencing, error correction models (ECM), or cointegration analysis (Engle-Granger or Johansen).
- **Omitted variables:** The model does not control for other determinants of inflation including global commodity prices, fiscal policy stance, output gap, exchange rate movements, or supply-side shocks. These omissions may bias the estimated relationship.
- **Annual data frequency:** Annual data masks significant within-year variation in both inflation and the repo rate. Quarterly or monthly data would allow more precise estimation of transmission lags.
- **Correlation vs. causation:** All results in this study are correlational. No causal claims are made. Establishing causality would require Granger causality tests or structural econometric models.

8 Conclusion

This study examined the relationship between CPI inflation and the RBI repo rate in India over the period 2007–2015 using correlation analysis, ADF stationarity testing, cross-correlation function analysis, and OLS lagged regression.

The key finding is that while contemporaneous correlation between the repo rate and inflation is negative (-0.61) — reflecting reactive monetary policy — the cross-correlation analysis reveals that the transmission effect of monetary policy peaks at a lag of 5–6 years. An OLS regression at this lag yields a directional R^2 of 0.803, though this must be treated cautiously given the small effective sample.

These findings reinforce the importance of lag-aware analysis in monetary policy evaluation. Contemporaneous measures understate or misrepresent the true nature of the relationship between policy instruments and inflation outcomes. For India specifically, the prolonged transmission lag is attributable to structural features of the banking system, supply-side inflation drivers, and the absence of a formal inflation targeting framework during the sample period.

Future research should extend this analysis using quarterly data over a longer time horizon (2000–2024), apply cointegration and Vector Autoregression (VAR) models to address non-stationarity and bidirectional causality, and incorporate additional macroeconomic controls to isolate the monetary policy channel.

References

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